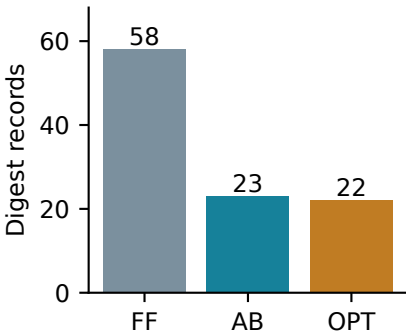
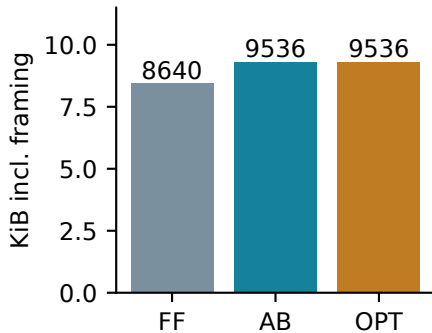


Capacity 23 → 22: unchanged online bytes, larger initialization

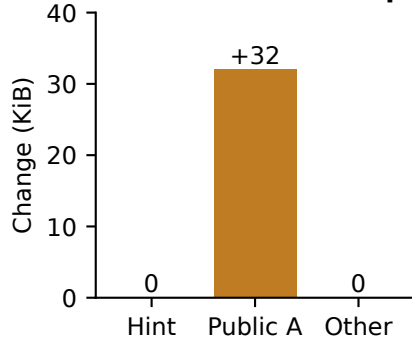
Largest bucket U



Online bytes / proof



OPT – AB bootstrap



Full-cache: 12,830 B bootstrap, 0 B online. OPT is a separate exact witness.